

QUESTION 1

Part 1

$$\text{Current yield} = \frac{\text{Annual Coupon}}{\text{Clean Price}}$$

Determine the dirty price

- ❖ Determining the dirty price.
 - The settlement date is 30th November while the books closed date for the next coupon is 27th November, so the bond is trading ex-coupon. **[2 marks]**
 - Yield to maturity(r) = $\frac{2.551\%}{2} = 1.2755$
 - The bond is left with two cashflows: one coupon and the face value.
 - The bond is trading ex-coupon, hence you discount the face value only

$$\text{All-In-Price} = \frac{R10,000}{(1.012755)^{\frac{183}{7}}} = R9,995.15 \text{ [8]}$$

Markers:

- ✓ **2 marks** for getting the 7 days correct.
 - ✓ **1 mark** for getting the 183 days correct.
 - ✓ **2 marks** for the correct semi-annual yield to maturity (YTM) of 1.2755%. If a student uses 2.551% then give **1 mark** (This is a mark for being able to pick the correct YTM from the snapshot- The YTM should be used or written in full with all its decimals).
 - ✓ **3 marks** for discounting the face value to the settlement date (**1 mark if a coupon is included**).
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- ❖ Determine the clean price.
 - ✓ Clean price = Dirty price + Accrued Interest
 - ✓ Accrued interest (Buyer) = $\frac{7}{365} \times 550 = 10.55$ **[3]**
 - **Marking note:** 2 marks for the correct coupon $\frac{5.5\% \times R10,000}{2} = R550$
 - If the coupon is wrong while everything else is correct, then 2 marks
 - ✓ Clean price = R9,995.15 + R10.55 = R10,005.70 **[3]**

$$\text{Current yield} = \frac{R550}{R10,005.70} = 5.497\% \text{ [3]}$$

Marking note: These are method marks

Part 2

The current yield is greater than the yield-to-maturity [1]. This is because the bond is trading at a premium, hence the yield-to-maturity accounts for built-in capital losses while the current yield captures the price only, without incorporating capital losses/gains [3]

Part 3

The yield-to-maturity assumes the bond is held to maturity while the realised compound yield works with any investment period [2].

Or The yield-to-maturity assumes the coupons are reinvested at the same interest rates as yield-to-maturity while the realised compound yield applies prevailing interest rates.

QUESTION 2

1. You position your portfolio by selling short-term bonds or short selling short-term bonds [1]. When the repo rate increases, yields of short-term bonds will increase and their prices drop. You maximise returns by buying-back short-term bonds at a lower price and make a profit equal to the difference between the selling price and buying-back price [3]
2. A repo rate cut means related interest rates like deposit and lending rates will also decrease [1]. A good example is the prime lending rate, which is benchmarked to the repo rate. The decrease in lending and deposit returns make bonds more attractive or more competitive compared to these other interest rate related instruments like lending and fixed deposits. This increases the demand for short term bonds; hence their prices will increase. As the bond prices increase, the yields will be decreasing – so a rate cut is likely to result in bond yields decreasing. [2].
3. SARB
 - a. The SARB left the repo rate unchanged at 8.25%. [1]. They believe the current repo rate restricts borrowing and it is in line with the inflation outlook [2]
Marking note: reasonable answers will be accepted.
 - b. They are likely to keep it unchanged at 8.25% in 2024 [1]. Given that their belief that their current repo rate is restrictive, they are unlikely to hike it. Given the elevated inflation expectations and upside risk they are also unlikely to cut the repo rate – hence they are likely to keep it unchanged in the next meeting. [2]
Marking note: reasonable answers will be accepted.
4. Reasonable and well justified answers will be accepted [1 Mark for stating the forecast and 2 marks for the justification]

QUESTION 3

Bond A is incorrectly quoted [1]. On the current quote the bond is trading at a discount. However, it is supposed to trade at a premium because its coupon rate is greater than the yield-to-maturity [3]

QUESTION 4

1. The traditional method is more likely to misprice bonds because it assumes a flat yield curve, which is rare. Which means there is chance of using the wrong discount rate and mispricing bonds if the yield curve is not flat. [2]. The arbitrage free valuation technique addresses this shortcoming by using the prevailing yield curve (term structure of interest) to discount each coupon. It views each coupon as a zero-coupon bond and then discounts each coupon using the yield to maturity of similar zero coupons bonds with the same maturity. [3]
 - *The arbitrage free valuation views a coupon bond as a package of zero-coupon bonds which can be split and sold separately. Therefore, when valuing a bond, it requires that each coupon be discounted using its own unique interest rate that correspond with the timing of each coupon [3]*
2. A recession is associated with a drop in interest rates [1] Investors will maximise returns by buying long-term bonds [1]. They will maximise your returns in two ways
 - It allows you to hold a long-term bond with a high coupon and you will continue receiving the high coupon even if interest rates eventually drop. [1]
 - When interest rates drop bond prices increase, hence you position your portfolio capital gains by buying long terms bonds which are more sensitive to interest rate changes than short term bonds. [2]
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QUESTION 5

1. Deriving forward rates

$$f_{(1*,1)} = \frac{(1+r)^T}{(1+r_{T-1})^{T-1}} - 1$$

$$r_{T-1} = \left(\frac{R1,000}{R934.58} \right) - 1 = 7\% [3]$$

To determine the forward rate in period 2 (starting 27th November 2024), you need the 2-year spot rate, it is not given, hence you need to bootstrap it using the 4-year coupon bond.

$$R1,069.58 = \frac{R100}{(1.07)^1} + \frac{R1,100}{(1+r)^2} [3]$$

$$r_2 = 6.16\%$$

$$f_{(1*,1)} = \left[\frac{(1+6.16\%)^2}{1+7\%} \right] - 1 = 5.32\% [3]$$

2. To infer future interest rates, inflation, to make a choice between long- and short-term bonds. 2 marks for any one of these

3. Arbitrage free value (you are valuing a corporate bond using a yield curve for government bonds, add spread to each spot rate)

$$P_b = \frac{R120}{(1+(7\%+1\%))^1} + \frac{R1,120}{(1+(6.16+1\%))^2} = R1,086.44 \text{ [4]}$$

Marking note:

- ✓ these are method marks, but the coupon payments should be correct. If they are wrong give 2 marks
- ✓ add a spread to the spot rate. Maximum 1 mark for those who don't add a spread.

4. If the market price is R850, then the bond is undervalued. You can buy the bond at R850 and then strip it and sell the cashflows separately as zero coupon bonds and realise a total of R1,105.94 [3]. Trading profit, R1,086.44-R850=236.44[2]

Marking note: The is a method mark, depending on the answer to the previous question.